

Local Exact Controllability of the Diffusion Equation in 1-D

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Abstract

This paper establishes the local exact null controllability of the diffusion equation in one dimension using distributed controls, in the case of the Dirichlet boundary value problem.

Most of the techniques used in the course of the proof are borrowed from [2].

1 Formulation of the problem

Consider the diffusion equation

$$(1.1) \quad \begin{aligned} y_t - (a(y))_{xx} &= mu & \forall (x, t) \in Q = I \times (0, T) \\ y(x, t) &= 0 & \forall (x, t) \in \Sigma = \partial I \times (0, T) \\ y(x, 0) &= y_0 & x \in I, \end{aligned}$$

where $I = (i_1, i_2)$ is a bounded real interval and m is the characteristic function of an open subset ω of I .

The ellipticity of this operator is insured by the condition $0 < \mu \leq a'(x)$, $\forall x \in \mathbb{R}$. We need $|a'(x)|$, $|a''(x)|$, and $|a'''(x)| \leq M < \infty$ as well, in order to establish the desired controllability result.

The equation (1.1) is said to be exactly null controllable if there is $u \in L^2(Q)$ such that $y(T) \equiv 0$, where $y \in C([0, T]; L^2(I))$ is the solution to (1.1).

We shall first establish some controllability results for the linearized equation

$$(1.2) \quad \begin{aligned} y_t - (a'(\tilde{y})y_x)_x &= mu & \forall (x, t) \in Q \\ y(x, t) &= 0 & \forall (x, t) \in \Sigma \\ y(x, 0) &= y_0 & x \in I, \end{aligned}$$

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where $\tilde{y}$ is some function with $\tilde{y}_x, \sqrt{t}\tilde{y}_t \in L^\infty(Q)$, $\tilde{y}_{xt} \in L^2(Q)$, and $\tilde{y}|_\Sigma = 0$. Henceforth, I shall denote $a'(\tilde{y})$ by b .

In order to establish the controllability of (1.2), it is sufficient to prove a Carleman-type inequality concerning the dual system

$$(1.3) \quad \begin{aligned} p_t + (bp_x)_x &= g & \forall (x, t) \in Q \\ p(x, t) &= 0 & \forall (x, t) \in \Sigma \\ p(x, T) &= p_T & x \in I. \end{aligned}$$

However, this only solves the problem for the linearized equation. Our purpose will be to eventually show that we can make $y = \tilde{y}$ in (1.2), since then the controllability of this equation would also imply that of the diffusion equation (1.1).

We shall arrive to the desired conclusion by means of the Kakutani fixed-point theorem and Pontryagin's principle. Indeed, under strong regularity conditions for the initial data ($(a(y_0))_{xx}$, $(y_0)_x$, and y_0 belonging to $L^2(I)$ and of sufficiently small norm), we shall see that the multifunction $y = \Phi(\tilde{y})$, where y is any solution to the equation (1.2), takes an $L^2(Q)$ -compact set into itself. The other conditions for applying Kakutani's theorem are easily met.

The final step will be to prove that we can dispense with at least some of the conditions on the initial data, due to the regularizing properties of the diffusion equation. We are going to show that, for every $y_0 \in H^1(I)$ of sufficiently small norm, the diffusion equation is null controllable.

2 The Carleman Inequality

In the following we shall assume the function a to be fixed, and therefore we shall not mention the constants μ and M explicitly. However, we shall keep track of $\tilde{y}$ and its derivatives.

In the course of the proof, we are going to need the following

Lemma 2.1 *There exists a function $\psi \in C^2(\bar{I})$ such that $\psi(x) > 0 \forall x \in I$, $\psi(x) = 0$ on ∂I , and $|\psi_x(x)| > 0 \forall x \in \bar{I} \setminus \omega_0$, where ω_0 is an open set such that $\overline{\omega_0} \subset I$.*

The proof is obvious (see [2] for a more general case).

Throughout the sequel we shall use a fixed ψ and a fixed $\overline{\omega_0} \subset \omega$.

In addition, let us define, for any $\lambda > 0$, the functions α and $\phi : Q \rightarrow \mathbb{R}$ by

$$\alpha(x, t) = \frac{e^{\lambda\psi(x)} - e^{2\lambda\|\psi\|_{C(\bar{I})}}}{t(T-t)}, \quad \phi(x, t) = \frac{e^{\lambda\psi(x)}}{t(T-t)}.$$

Note that $\phi(x, t) \geq c > 0$, $\forall (x, t) \in Q$, and $e^{\delta\alpha}\phi^k \leq C < \infty$, $\forall \delta > 0$, $k \in \mathbb{R}$.

We are now ready to formulate the Carleman inequality concerning equation (1.3). The proof below is, many ways, identical to the one given by O. Yu. Imanuvilov in a different case (see [7]).

Theorem 2.1 *For any solution p of the dual system (1.3), with $\|b_x\|_{L^\infty(Q)}$, $\|\sqrt{t}b_t\|_{L^\infty(Q)} \leq \rho$ and for every $\lambda \geq \lambda_0(\rho)$, $s \geq s_0(\lambda)$, the following inequality takes place:*

$$(2.1) \quad \int_Q e^{2s\alpha} (s^3 \phi^3 p^2 + s \phi p_x^2 + s^{-1} \phi^{-1} (p_t^2 + p_{xx}^2)) \leq \\ \leq C(\lambda, \rho) \left(\int_{Q_\omega} e^{2s\alpha} s^3 \phi^3 p^2 + \int_Q e^{2s\alpha} g^2 \right),$$

where $C(\lambda, \rho)$ is a constant independent of p , g , and s , but which may depend on ψ , λ , and ρ , and $Q_\omega = \omega \times (0, T)$.

Proof of Theorem 2.1 This proof follows step by step the one given in [2] for Theorem 1.2.1 (the Carleman inequality concerning the heat equation, pp. 145 - 152).

By taking $z = e^{2s\alpha} p$ in the dual system (1.3) we obtain

$$(2.2) \quad \begin{aligned} & z_t - s\alpha_t z + (bz_x)_x - 2s\lambda\phi\psi_x \cdot bz_x + \\ & + (s^2\lambda^2\phi^2\psi_x^2 - s\lambda^2\phi\psi_x^2 - s\lambda\phi\psi_{xx})bz - s\lambda\phi\psi_x b_x z = e^{s\alpha} g \\ & z(0) = z(T) = 0 \text{ on } I \\ & z|_{\Sigma} = 0. \end{aligned}$$

If we set

$$\begin{aligned} B(t)z &= -(bz_x)_x - (s^2\lambda^2\phi^2\psi_x^2 + s\lambda^2\phi\psi_x^2)bz + s\alpha_t z \\ X(t)z &= -2s\lambda^2\phi\psi_x^2 bz - 2s\lambda\phi\psi_x \cdot bz_x \\ Z(t)z &= s\lambda\phi\psi_x b_x z + s\lambda\phi\psi_{xx} bz \end{aligned}$$

the equation can be rewritten as $z_t - B(t)z + X(t)z = Z(t)z + e^{s\alpha} g$. Then, starting with the relation

$$\begin{aligned} \frac{d}{dt} \int_I B(t)z \cdot z \, dx &= \int_I B(t)z_t \cdot z + B(t)z \cdot z_t + B_t(t)z \cdot z \, dx \\ &= 2 \int_I B(t)z (B(t)z - X(t)z + Z(t)z + e^{s\alpha} g) \, dx + \int_I B_t(t)z \cdot z \, dx \end{aligned}$$

and integrating it on $(0, T)$ we obtain

$$(2.3) \quad 2 \int_Q (B(t)z)^2 + 2Y = -2 \int_Q B(t)z (Z(t)z + e^{s\alpha} g) - \int_Q B_t(t)z \cdot z,$$

where $Y = -\int_Q B \cdot X$, that is

$$(2.4) \quad Y = -\int_Q (2s\lambda\phi\psi_x \cdot bz_x + 2s\lambda^2\phi\psi_x^2bz) \cdot ((bz_x)_x + (s^2\lambda^2\phi^2\psi_x^2 + s\lambda^2\phi\psi_x^2)bz - s\alpha_t z).$$

Then, let us evaluate

$$\int_Q B_t(t)z \cdot z = \int_Q b_t z_x^2 - \int_Q (s^2\lambda^2\phi^2\psi_x^2b + s\lambda^2\phi\psi_x^2b)_t z^2 + s\alpha_{tt} z^2.$$

Set $\gamma(\lambda) = e^{2\lambda\|\psi\|_{C(\bar{T})}}$ and let us take $s \geq \gamma(\lambda)$, $\lambda \geq \lambda_0$. In addition, put

$$D(s, \lambda, z) = \int_Q s^3\lambda^3\phi^3z^2 + s\lambda\phi z_x^2.$$

We eventually obtain that

$$(2.5) \quad \left| \int_Q B_t(t)z \cdot z \right| \leq C(1 + \|\phi^{-1}b_t\|_{L^\infty(Q)})D(s, \lambda, z).$$

Furthermore,

$$(2.6) \quad 2 \left| \int_Q B(t)z(Z(t)z + e^{s\alpha}g) \right| \leq \int_Q (B(t)z)^2 + C(1 + \|\phi^{-1}b_x^2\|_{L^\infty(Q)})D(s, \lambda, z) + C \int_Q e^{2s\alpha}g^2.$$

From (2.3), (2.5), and (2.6) we obtain

$$(2.7) \quad Y \leq C(1 + \|\sqrt{t}b_t\|_{L^\infty(Q)} + \|b_x\|_{L^\infty(Q)}^2)D(s, \lambda, z) + C \int_Q e^{2s\alpha}g^2.$$

On the other hand, we have the following lower estimates:

$$(2.8) \quad \left| \int_Q (2s\lambda\phi\psi_x \cdot bz_x + s\lambda\phi\psi_x b_x z) s\alpha_t z \right| \leq CD(s, \lambda, z)$$

for $s \geq \gamma(\lambda)$; then,

$$(2.9) \quad \left| \int_Q s\lambda\phi\psi_x b_x z \cdot s\alpha_t z \right| \leq C\|b_x\|_{L^\infty(Q)} \cdot D(s, \lambda, z),$$

and

$$(2.10) \quad \left| \int_Q (s\lambda^2 \phi \psi_x^2 b z) s \alpha_t z \right| \leq CD(s, \lambda, z).$$

Moreover,

$$(2.11) \quad \begin{aligned} & - \int_Q s\lambda^2 \phi \psi_x^2 b z \cdot (bz_x)_x = \int_Q s\lambda^2 (\phi \psi_x^2 b z)_x \cdot bz_x \geq \\ & \geq \int_Q s\lambda^2 \phi \psi_x^2 b^2 z_x^2 - C(1 + \|b_x\|_{L^\infty(Q)}^2) D(s, \lambda, z). \end{aligned}$$

Furthermore,

$$(2.12) \quad \begin{aligned} & - \int_Q (2s\lambda \phi \psi_x \cdot bz_x) (s^2 \lambda^2 \phi^2 \psi_x^2 + s\lambda^2 \phi \psi_x^2) b z = \\ & = \int_Q (s^3 \lambda^3 \phi^3 \psi_x^3 b^2 + s^2 \lambda^3 \phi^2 \psi_x^3 b^2)_x \cdot z^2 \geq \\ & \geq \int_Q (3s^3 \lambda^4 \phi^3 \psi_x^4 + 2s^2 \lambda^4 \phi^2 \psi_x^4) \cdot b^2 z^2 - C(1 + \|b_x\|_{L^\infty(Q)}) D(s, \lambda, z). \end{aligned}$$

Finally,

$$(2.13) \quad - \int_Q 2s\lambda \phi \psi_x \cdot bz_x (bz_x)_x \geq - \int_\Sigma s\lambda \phi \frac{d\psi}{d\nu} \cdot b^2 z_x^2 d\sigma - CD(s, \lambda, \phi).$$

Since $\psi > 0$ on I and $\psi = 0$ on ∂I , we have $\frac{d\psi}{d\nu} \leq 0$, and therefore

$$- \int_\Sigma s\lambda \phi \frac{d\psi}{d\nu} \cdot b^2 z_x^2 d\sigma \geq 0.$$

Combining the inequalities (2.8), (2.9), (2.10), (2.11), (2.12), and (2.13), we obtain

$$(2.14) \quad Y \geq \int_Q s^3 \lambda^4 \phi^3 \psi_x^4 b^2 z^2 + s\lambda^2 \phi \psi_x^2 b^2 z^2 - C(1 + \|b_x\|_{L^\infty(Q)}^2) D(s, \lambda, z)$$

From (2.14) and (2.7) we get

$$\begin{aligned}
 & \int_Q s^3 \lambda^4 \phi^3 \psi_x^4 b^2 z^2 + s \lambda^2 \phi \psi_x^2 b^2 z^2 \leq \\
 (2.15) \quad & \leq C(1 + \|\sqrt{t}b_t\|_{L^\infty(Q)} + \|b_x\|_{L^\infty(Q)}^2)D(s, \lambda, z) + C \int_Q e^{2s\alpha} g^2 \\
 & \leq C(\rho)D(s, \lambda, z) + C \int_Q e^{2s\alpha} g^2.
 \end{aligned}$$

Because $b \geq \mu$ and $|\psi_x| \geq c$ on $I \setminus \omega_0$, by making λ sufficiently large ($\lambda \geq 2C(\rho)$, for instance) it follows that

$$\begin{aligned}
 & \int_Q s^3 \lambda^4 \phi^3 z^2 + s \lambda^2 \phi \psi_x^2 b^2 z^2 \leq \\
 (2.16) \quad & \leq C(\rho) \int_{Q_{\omega_0}} s^3 \lambda^4 \phi^3 z^2 + s \lambda^2 \phi \psi_x^2 b^2 z^2 + C \int_Q e^{2s\alpha} g^2
 \end{aligned}$$

Hence we can obtain, by exactly the methods used in [2] (p. 151), that

$$\begin{aligned}
 & \int_Q e^{2s\alpha} s^3 \lambda^4 \phi^3 p^2 + s \lambda^2 \phi p_x^2 \leq \\
 (2.17) \quad & \leq C(\rho) \int_{Q_{\omega_0}} e^{2s\alpha} s^3 \lambda^4 \phi^3 p^2 + s \lambda^2 \phi p_x^2 + C \int_Q e^{2s\alpha} g^2.
 \end{aligned}$$

Choose $\chi \in C_0^\infty(\omega)$ such that $\chi = 1$ in ω_0 . If we multiply equation (1.3) by $e^{2s\alpha} \chi \phi p$ and integrate on Q we obtain that

$$\begin{aligned}
 & \int_Q e^{2s\alpha} \chi \phi b p_x^2 = - \int_Q (e^{2s\alpha} \chi \phi)_x \cdot b p p_x + \int_Q e^{2s\alpha} \chi \phi p g + \int_Q \chi (e^{2s\alpha} \phi)_t p^2 \leq \\
 & \leq C(1 + \|b_x\|_{L^\infty(Q)}) \int_{Q_\omega} e^{2s\alpha} s^2 \lambda^2 \phi^3 z^2 + C \int_Q e^{2s\alpha} g^2.
 \end{aligned}$$

Consequently we get

$$\int_Q e^{2s\alpha} s^3 \lambda^4 \phi^3 p^2 + s \lambda^2 \phi p_x^2 \leq C(\rho) \int_{Q_\omega} e^{2s\alpha} s^3 \lambda^4 \phi^3 p^2 + C \int_Q e^{2s\alpha} g^2.$$

Equivalently, we may say that, for $\lambda \geq \lambda_0(\rho)$, $s \geq s_0(\lambda)$,

$$(2.18) \quad \int_Q e^{2s\alpha} s^3 \phi^3 p^2 + s \phi p_x^2 \leq C(\rho, \lambda) \int_{Q_\omega} e^{2s\alpha} s^3 \phi^3 p^2 + C \int_Q e^{2s\alpha} g^2.$$

Next, by squaring equation (1.3), multiplying it with $e^{2s\alpha}s^{-1}\phi^{-1}$, and integrating on Q , we obtain

$$(2.19) \quad \int_Q e^{2s\alpha}s^{-1}\phi^{-1}(p_t + (bp_x)_x)^2 \leq C \int_Q e^{2s\alpha}g^2.$$

At the same time,

$$(2.20) \quad \begin{aligned} & \int_Q e^{2s\alpha}s^{-1}\phi^{-1}p_t \cdot (bp_x)_x = - \int_Q (e^{2s\alpha}\phi^{-1}p_t)_x \cdot bp_x \geq \\ & \geq \frac{1}{2} \int_Q (e^{2s\alpha}\phi^{-1})_t \cdot p_x^2 - \frac{1}{2} \int_Q e^{2s\alpha}s^{-1}\phi^{-1}p_t^2 - C \int_Q s\lambda^2\phi p_x^2 \geq \\ & \geq -\frac{1}{2} \int_Q e^{2s\alpha}s^{-1}\phi^{-1}p_t^2 - C \int_Q s\lambda^2\phi p_x^2. \end{aligned}$$

By (2.19) and (2.20) we obtain

$$(2.21) \quad \int_Q e^{2s\alpha}s^{-1}\phi^{-1}(p_t^2 + (bp_x)_x^2) \leq C \int_Q e^{2s\alpha}g^2 + C \int_Q s\lambda^2\phi p_x^2.$$

Combining inequalities (2.18) and (2.21) we obtain the final result

$$\begin{aligned} & \int_Q e^{2s\alpha}(s^3\phi^3p^2 + s\phi p_x^2 + s^{-1}\phi^{-1}(p_t^2 + p_{xx}^2)) \leq \\ & \leq C(\lambda, \rho) \left(\int_{Q_\omega} e^{2s\alpha}s^3\phi^3p^2 + \int_Q e^{2s\alpha}g^2 \right). \end{aligned}$$

Proceeding as in [2] (see Corollary 1.2.1, p. 145), one may obtain the following corollary:

Corollary 2.1 *Under the assumptions of Theorem 2.1, the following inequality holds:*

$$(2.22) \quad \int_I p^2(0) dx \leq C(s, \lambda, \rho) \left(\int_{Q_\omega} e^{2s\alpha}s^3\phi^3p^2 + \int_Q e^{2s\alpha}g^2 \right),$$

where $C(s, \lambda, \rho)$ is a constant that does not depend on p or g .

3 Estimates for y

First, let us denote $Q_t = I \times (0, t)$ and $f = mu$. Thus, the system (1.2) can be rewritten

$$(3.1) \quad \begin{aligned} y_t - (by_x)_x &= f & \forall (x, t) \in Q \\ y(x, t) &= 0 & \forall (x, t) \in \Sigma \\ y(x, 0) &= y_0 & x \in I, \end{aligned}$$

We shall prove a few estimates concerning the solution y of the system (3.1).

Multiplying equation (3.1) by y_t and integrating on Q_t we get

$$\int_{Q_t} y_t^2 + \int_{Q_t} \frac{d}{dt} b y_x^2(t) \leq C \left(\int_{Q_t} f^2 + \int_{Q_t} b_t y_x^2 \right),$$

so

$$\begin{aligned} & \int_Q y_t^2 + \sup_{t \in [0, T]} \int_I y_x^2(t) dx \leq \\ (3.2) \quad & \leq C \left(\int_I y_x^2(0) dx + \int_Q f^2 + \int_0^T \frac{dt}{\sqrt{t}} \cdot \|\sqrt{t} b_t\|_{L^\infty(Q)} \cdot \sup_{t \in [0, T]} \int_I y_x^2(t) dx \right) \\ & \leq C_0 \left(\int_I y_x^2(0) dx + \int_Q f^2 + \|\sqrt{t} b_t\|_{L^\infty(Q)} \cdot \sup_{t \in [0, T]} \int_I y_x^2(t) dx \right). \end{aligned}$$

Making $\|\sqrt{t} b_t\|_{L^\infty(Q)}$ sufficiently small ($\|\sqrt{t} b_t\|_{L^\infty(Q)} \leq \frac{1}{2C_0}$), from (3.2) we infer that $y \in L^\infty((0, T); H_0^1(I)) \subset L^\infty(Q)$, and

$$(3.3) \quad \|y\|_{L^\infty(Q)}^2 \leq C \sup_{t \in [0, T]} \int_I y_x^2(t) dx \leq C \left(\int_I y_x^2(0) dx + \int_Q f^2 \right).$$

As a consequence of the fact that $(by_x)_x = y_t - f$, we also obtain from (3.3) that

$$(3.4) \quad \int_Q (by_x)_x^2 \leq C \left(\int_I y_x^2(0) dx + \int_Q f^2 \right).$$

Multiplying equation (3.1) by $(by_x)_{xt}$ and integrating on Q_t we get

$$\int_{Q_t} y_t (by_x)_{xt} - \int_{Q_t} (by_x)_x (by_x)_{xt} = \int_{Q_t} f (by_x)_{xt},$$

and by Green's formula

$$(3.5) \quad \int_{Q_t} (by_{xt}^2 + b_t y_x y_{xt}) + \frac{1}{2} \int_{Q_t} \frac{d}{dt} (by_x)_x^2 = - \int_{Q_t} f (by_x)_{xt}.$$

Through integration by parts we obtain

$$\begin{aligned} & - \int_{Q_t} f (by_x)_{xt} = - \int_I f (by_x)_x dx \Big|_0^t + \int_{Q_t} f_t (by_x)_x \leq \\ & \leq \int_I (f^2(0) + f^2(t)) dx + \frac{1}{4} \int_I ((by_x)_x^2(0) + (by_x)_x^2(t)) dx + \int_{Q_t} (by_x)_x^2 + \int_{Q_t} f_t^2 \\ & \leq \frac{1}{4} \int_I ((by_x)_x^2(0) + (by_x)_x^2(t)) dx + \int_{Q_t} (by_x)_x^2 + \int_{Q_t} (f_t^2 + f^2). \end{aligned}$$

An elementary computation also proves that

$$\begin{aligned} \int_{Q_t} y_{xt} b_t y_x &\leq \frac{1}{4c} \int_{Q_t} y_{xt}^2 + c \int_0^t \|b_t(s)\|_{L^\infty(I)}^2 ds \cdot \sup_{s \in [0, T]} \int_I y_x^2 dx \\ &\leq \frac{1}{4c} \int_{Q_t} y_{xt}^2 + C \|b_{xt}\|_{L^2(Q)}^2 \cdot \sup_{s \in [0, t]} \int_I y_x^2 dx \end{aligned}$$

Applying the preceding estimates into the equality (3.5) gives us

$$\begin{aligned} (3.6) \quad \int_{Q_t} b y_{xt}^2 + \int_I (b y_x)_x^2(t) dx &\leq C \left(\int_I (b y_x)_x^2(0) dx + \int_{Q_t} (b y_x)_x^2 + \right. \\ &\quad \left. + \int_{Q_t} (f_t^2 + f^2) + \|b_{xt}\|_{L^2(Q)}^2 \cdot \sup_{s \in [0, t]} \int_I y_x^2(0) dx \right). \end{aligned}$$

Taking into account (3.3) and (3.4), for any $t \in [0, T]$ it is true that

$$\begin{aligned} (3.7) \quad \int_Q y_{xt}^2 + \int_I (b y_x)_x^2(t) dx &\leq C \left(\int_I (b y_x)_x^2(0) dx + \int_Q f_t^2 + \right. \\ &\quad \left. + (1 + \|b_{xt}\|_{L^2(Q)}^2) \left(\int_I y_x^2(0) dx + \int_Q f^2 \right) \right). \end{aligned}$$

One consequence of this relation and of (3.3) is that $b y_x \in L^\infty((0, T); H^1(I)) \subset L^\infty(Q)$, and

$$\begin{aligned} (3.8) \quad \|y_x\|_{L^\infty(Q)}^2 &\leq C \left(\int_I (b y_x)_x^2(0) dx + \int_Q f_t^2 + \right. \\ &\quad \left. + (1 + \|b_{xt}\|_{L^2(Q)}^2) \left(\int_I y_x^2(0) dx + \int_Q f^2 \right) \right). \end{aligned}$$

Taking the derivative of equation (3.1) with respect to t , we obtain

$$y_{tt} - (b y_{tx})_x - (b_t y_x)_x = f_t.$$

Multiplying this with $t y_{tt}$ and integrating on Q_t we have

$$\int_{Q_t} t y_{tt}^2 + \int_{Q_t} t b y_{xt} y_{xtt} = \int_{Q_t} t ((b_t y_x)_x + f_t) y_{tt};$$

it follows that

$$\int_{Q_t} ty_{tt}^2 + \frac{1}{2} \int_{Q_t} \frac{d}{dt}(tb y_{xt}^2) \leq C \left(\int_{Q_t} t f_t^2 + t(b_t y_x)_x^2 + \int_{Q_t} (tb_t + b) y_{xt}^2 \right).$$

Then, since $(b_t y_x)_x = b_{tx} y_x + b_t y_{xx}$,

$$\begin{aligned} \int_{Q_t} t(b_t y_x)_x^2 &\leq 2 \left(\int_Q t b_{tx}^2 y_x^2 + \int_Q t b_t^2 y_{xx}^2 \right) \\ &\leq C \|b_{xt}\|_{L^2(Q)}^2 \cdot \|y_x\|_{L^\infty(Q)}^2 + \|\sqrt{t} b_t\|_{L^\infty(Q)}^2 \cdot \int_Q y_{xx}^2. \end{aligned}$$

Furthermore, it is clearly true that

$$\begin{aligned} \int_{Q_t} t b_t y_{xt}^2 &\leq \int_0^t \frac{ds}{\sqrt{s}} \cdot \|\sqrt{t} b_t\|_{L^\infty(Q)} \cdot \sup_{s \in [0, t]} \int_I s y_{xt}^2(s) dx \\ &\leq C \|\sqrt{t} b_t\|_{L^\infty(Q)} \cdot \sup_{t \in [0, T]} \int_I t y_{xt}^2(t) dx. \end{aligned}$$

It follows that

$$\begin{aligned} (3.9) \quad \int_Q y_{tt}^2 + \sup_{t \in [0, T]} \int_I t y_{xt}^2(t) dx &\leq C_1 \left(\int_Q f_t^2 + \|y_x\|_{L^\infty(Q)}^2 \cdot \|b_{xt}\|_{L^2(Q)}^2 + \right. \\ &\quad \left. + \|\sqrt{t} b_t\|_{L^\infty(Q)}^2 \int_Q y_{xx}^2 + \int_Q y_{xt}^2 + \|\sqrt{t} b_t\|_{L^\infty(Q)} \cdot \sup_{t \in [0, T]} \int_I t y_{xt}^2(t) dx \right). \end{aligned}$$

Since $b y_{xx} = (b y_x)_x - b_x y_x$, from (3.3) and (3.4) we can infer that

$$\int_Q y_{xx}^2 \leq C(1 + \|b_x\|_{L^\infty(Q)}^2) \left(\int_I y_x^2(0) dx + \int_Q f^2 \right).$$

Making this substitution, as well as those possible in virtue of (3.8) and (3.7), into (3.9), we get (for every b with $\|\sqrt{t} b_t\|_{L^\infty(Q)} \leq \frac{1}{2C_1}$)

$$\begin{aligned} \int_Q y_{tt}^2 + \sup_{t \in [0, T]} \int_I t y_{xt}^2(t) dx &\leq C \left(\int_Q f_t^2 + \right. \\ &\quad \left. + (\|b_{xt}\|_{L^2(Q)}^2 + \|\sqrt{t} b_t\|_{L^\infty(Q)}^2 (1 + \|b_x\|_{L^\infty(Q)}^2)) \left(\int_I y_x^2(0) dx + \int_Q f^2 \right) + \right. \\ &\quad \left. + \int_I (b y_x)_x^2(0) dx + \int_Q f_t^2 + (1 + \|b_{xt}\|_{L^2(Q)}^2) \left(\int_I y_x^2(0) dx + \int_Q f^2 \right) \right), \end{aligned}$$

which becomes (after sorting out the terms)

$$\begin{aligned} \int_Q y_{tt}^2 + \sup_{t \in [0, T]} \int_I t y_{xt}^2(t) \, dx &\leq C \left(\int_Q f_t^2 + \int_I (b y_x)_x^2(0) \, dx + \right. \\ &\left. + (1 + \|b_{xt}\|_{L^2(Q)}^2 + \|\sqrt{t} b_t\|_{L^\infty(Q)}^2 (1 + \|b_x\|_{L^\infty(Q)}^2)) \left(\int_I y_x^2(0) \, dx + \int_Q f^2 \right) \right). \end{aligned}$$

As a consequence, we see that $\sqrt{t} y_t \in L^\infty((0, T); H_0^1(I)) \subset L^\infty(Q)$ (with $\|\sqrt{t} y_t\|_{L^\infty(Q)} \leq C \|y_{xt}\|_{L^\infty((0, T); L^2(I))}$).

Coming back to $\tilde{y}$ (remember that $b = a'(\tilde{y})$), let us note that

$$\|b_x\|_{L^\infty(Q)} \leq C \|\tilde{y}_x\|_{L^\infty(Q)}, \quad \|\sqrt{t} b_t\|_{L^\infty(Q)} \leq C \|\sqrt{t} \tilde{y}_t\|_{L^\infty(Q)},$$

and

$$\begin{aligned} \|b_{xt}\|_{L^2(Q)} &\leq C (\|\tilde{y}_x\|_{L^\infty(Q)} \cdot \|\tilde{y}_t\|_{L^2(Q)} + \|\tilde{y}_{xt}\|_{L^2(Q)}) \\ &\leq C \|\tilde{y}_x\|_{L^\infty(Q)} (1 + \|\tilde{y}_{xt}\|_{L^2(Q)}). \end{aligned}$$

Let us also remember that $f = mu$.

By a very rough estimate we obtain

$$\begin{aligned} (3.10) \quad \|y_x\|_{L^\infty(Q)}^2 + \|y_{xt}\|_{L^2(Q)}^2 &\leq C(1 + \|\tilde{y}_x\|_{L^\infty(Q)}^2 (1 + \|\tilde{y}_{xt}\|_{L^2(Q)}^2)) \\ &\quad \left(\int_I (b y_x)_x^2(0) + y_x^2(0) \, dx + \int_Q u_t^2 + u^2 \right) \end{aligned}$$

and

$$\begin{aligned} (3.11) \quad \|\sqrt{t} y_t\|_{L^\infty(Q)}^2 &\leq C(1 + \|\sqrt{t} \tilde{y}_t\|_{L^\infty(Q)}^2 (1 + \|\tilde{y}_x\|_{L^\infty(Q)}^2 (1 + \|\tilde{y}_{xt}\|_{L^2(Q)}^2))) \\ &\quad \left(\int_I (b y_x)_x^2(0) + y_x^2(0) \, dx + \int_Q u_t^2 + u^2 \right), \end{aligned}$$

for all $\tilde{y}$ with $\|\sqrt{t} \tilde{y}_t\|_{L^\infty(Q)} \leq C_2 = \frac{1}{\mu} \max(\frac{1}{2C_0}, \frac{1}{2C_1})$, where C_2 is a constant which does not depend on $\tilde{y}$, u , or y .

Further compressing these results, we arrive at

$$\begin{aligned} (3.12) \quad \|y_x\|_{L^\infty(Q)}^2 + \|\sqrt{t} y_t\|_{L^\infty(Q)}^2 + \|y_{xt}\|_{L^2(Q)}^2 &\leq C(1 + \|\sqrt{t} \tilde{y}_t\|_{L^\infty(Q)}^2) \cdot \\ &\quad (1 + \|\tilde{y}_x\|_{L^\infty(Q)}^4 + \|\tilde{y}_{xt}\|_{L^2(Q)}^4) \left(\int_I (b y_x)_x^2(0) + y_x^2(0) \, dx + \int_Q u_t^2 + u^2 \right). \end{aligned}$$

4 Optimization and the Main Result

Now we are ready to establish the main theorem, first for y_0 in a narrower class of functions, then for $y_0 \in H^1(I)$.

Theorem 4.1 *For any $\delta > 0$, there exists $\eta > 0$ such that, for every $y_0 \in L^2(I)$ with $\|(a(y_0))_{xx}\|_{L^2(I)} + \|(y_0)_x\|_{L^2(I)} + \|y_0\|_{L^2(I)} \leq \eta$, the equation (1.1) is exactly null controllable, with a controller u satisfying*

$$\int_Q e^{2s(\delta-1)\alpha} (u^2 + u_t^2) \leq C_3(\delta) \|y_0\|_{L^2}^2.$$

Proof of Theorem 4.1 Let us define

$$K = \{y \mid \|y_x\|_{L^\infty(Q)} \leq \rho, \|\sqrt{t}y_t\|_{L^\infty(Q)} \leq \rho, \|y_{xt}\|_{L^2(Q)} \leq \rho, y(0) = y_0\}.$$

Note that this set is compact in the $L^2(Q)$ topology. Indeed, it can be easily seen that the set is closed, and the fact that $\|y\|_{H^1(Q)}$ is bounded insures its precompactness.

Consider the linearized equation (1.2) for $\tilde{y} \in K$. By the definition of K , for all $\tilde{y} \in K$ we have a fixed ρ (which does not depend on $\tilde{y}$) in the Carleman inequality (Theorem 2.1). Setting a sufficiently small ρ ($\rho \leq C_4$), we also find that we can apply the main result of the previous section, (3.12), for all the elements of K .

For brevity, in the following we shall denote any constant which does not depend on $p, y, \tilde{y}, g$, and u by C (such a constant may, however, depend on s, λ, ρ , or δ).

Consider the optimal control problem: minimize

$$(4.1) \quad \int_Q e^{-2s\alpha} \phi^{-3} u^2 + \epsilon^{-1} \int_I e^{-2s\alpha} y^2(T) dx$$

subject to (1.2). By the Pontryagin principle this problem has a unique solution (u_ϵ, y_ϵ) . We have $u_\epsilon = mp_\epsilon e^{2s\alpha} \phi^3$, where p is a solution of the dual system

$$(4.2) \quad \begin{aligned} (p_\epsilon)_t + (b \cdot (p_\epsilon)_x)_x &= 0 & \forall (x, t) \in Q \\ p_\epsilon(x, t) &= 0 & \forall (x, t) \in \Sigma \\ p_\epsilon(x, T) &= -\frac{1}{\epsilon} y_\epsilon(x, T) & \forall x \in I. \end{aligned}$$

Multiplying (4.2) by y_ϵ , (1.2) by p_ϵ , adding the two equations together and integrating on Q we get

$$(4.3) \quad \begin{aligned} \int_{Q_\omega} e^{2s\alpha} \phi^3 p_\epsilon^2 + \epsilon^{-1} \int_I e^{-2s\alpha} y_\epsilon^2(T) dx &= - \int_I y_0 p(0) dx \\ &\leq \frac{1}{2k} \int_I y_0^2 dx + \frac{k}{2} \int_I p(0)^2 dx \end{aligned}$$

Applying Corollary (2.1), and choosing a small enough k in the inequality above, we further obtain

$$(4.4) \quad \int_{Q_\omega} e^{2s\alpha} \phi^3 p_\epsilon^2 + \epsilon^{-1} \int_I e^{-2s\alpha} y_\epsilon^2(T) dx \leq C \int_I y_0^2 dx$$

Clearly

$$\int_Q e^{2s(\delta-1)\alpha} u_\epsilon^2 = \int_{Q_\omega} e^{2s\alpha} \phi^3 p_\epsilon^2 \cdot e^{2s\delta\alpha} \phi^3 \leq C \int_{Q_\omega} e^{2s\alpha} \phi^3 p_\epsilon^2$$

and

$$\begin{aligned} \int_Q e^{2s(\delta-1)\alpha} (u_\epsilon)_t^2 &= \int_{Q_\omega} e^{2s(\delta-1)\alpha} (e^{2s\alpha} \phi^3 (p_\epsilon)_t + 2se^{2s\alpha} \alpha_t \phi^3 p_\epsilon + 3e^{2s\alpha} \phi^2 \phi_t p_\epsilon)^2 \\ &\leq C \left(\int_Q e^{2s\alpha} \phi^{-1} (p_\epsilon)_t^2 \cdot e^{2s\delta\alpha} \phi^7 + \int_{Q_\omega} e^{2s\alpha} \phi^3 p_\epsilon^2 \cdot e^{2s\delta\alpha} \phi^7 \right) \\ &\leq C \int_{Q_\omega} e^{2s\alpha} \phi^3 p_\epsilon^2. \end{aligned}$$

Combining these last relations and (4.4) gives

$$(4.5) \quad \int_Q e^{2s(\delta-1)\alpha} (u_\epsilon^2 + (u_\epsilon)_t^2) + \epsilon^{-1} \int_I e^{-2s\alpha} y_\epsilon^2(T) dx \leq C_3 \int_I y_0^2 dx.$$

Taking $\delta = 1$ and applying (3.12) (and also considering that $\|\tilde{y}_x\|_{L^\infty(Q)} \leq \rho$, $\|\sqrt{t}\tilde{y}_t\|_{L^\infty(Q)} \leq \rho$, $\|\tilde{y}_{xt}\|_{L^2(Q)} \leq \rho$) gives

$$\begin{aligned} &\|y_x\|_{L^\infty(Q)}^2 + \|\sqrt{t}y_t\|_{L^\infty(Q)}^2 + \|y_{xt}\|_{L^2(Q)}^2 \leq \\ &\leq C(1 + \rho^2)(1 + 2\rho^4) \left(\int_I (by_x)_x^2(0) + y_x^2(0) dx + \int_Q (u_\epsilon)_t^2 + u_\epsilon^2 \right) \\ &\leq C_4(1 + \rho^6) \int_I (b \cdot (y_0)_x)_x^2 + (y_0)_x^2 + y_0^2 dx \end{aligned}$$

Furthermore, since $b(0) = a'(\tilde{y}(0)) = a'(y_0)$, we obtain

$$\int_I (by_x)_x^2(0) + y_x^2(0) + y_0^2 dx = \int_I (a(y_0))_{xx}^2(0) + (y_0)_x^2 + y_0^2 dx,$$

hence the condition imposed on y_0 in the statement of the theorem. Thus we get

$$(4.6) \quad \begin{aligned} \|y_x\|_{L^\infty(Q)}^2 + \|\sqrt{t}y_t\|_{L^\infty(Q)}^2 + \|y_{xt}\|_{L^2(Q)}^2 &\leq \\ &\leq C_4(1 + \rho^6) \int_I (a(y_0))_{xx}^2(0) + (y_0)_x^2 + y_0^2 dx. \end{aligned}$$

By making

$$\int_I (a(y_0))_{xx}^2 + (y_0)_x^2 + y_0^2 dx < \eta$$

for a sufficiently small η (such as $\eta = \frac{\rho}{C_4(1+\rho^6)}$), we obtain that $y_\epsilon \in K$.

This means that u_ϵ and $(u_\epsilon)_t$ belong to a bounded set in $L^2(Q)$, the same for every ϵ , and y_ϵ belongs to K , set that is compact in $L^2(Q)$.

Choose a sequence of (u_ϵ, y_ϵ) , $\epsilon \rightarrow 0$, that achieve the optimum in expression (4.1). From the above it follows that (on a subsequence)

$$(4.7) \quad \begin{aligned} u_\epsilon &\rightarrow u && \text{weakly in } L^2(Q) \\ (u_\epsilon)_t &\rightarrow v && \text{weakly in } L^2(Q) \\ y_\epsilon &\rightarrow y && \text{strongly in } L^2(Q). \end{aligned}$$

Obviously it follows that $v = u_t$. Other immediate consequences are that $(y_\epsilon)_t$ tends to y_t in $H^{-1}(Q)$ and that $(b(y_\epsilon)_x)_x$ tends to $(by_x)_x$ in $L^2((0, T); H^{-2}(I))$; then (u, y) satisfy the linearized equation (1.2). Furthermore, the strong convergence of y_ϵ implies that $y \in K$. Last, but not least, by making $\epsilon \rightarrow 0$ in (4.4) we see that $y(T) \equiv 0$.

As for $\|e^{s(1-\delta)\alpha}u\|_{L^2(Q)}$ and $\|e^{s(1-\delta)\alpha}u_t\|_{L^2(Q)}$, let us remember that, for any sequence $(f_n)_{n \in \mathbb{N}}$ that converges weakly to f in L^2 , we have $\|f\|_{L^2(Q)} \leq \liminf \|f_n\|_{L^2(Q)}$.

Thus, we have obtained a pair (y, u) which satisfies the system (1.2), such that $y \in K$, $y(T) \equiv 0$, and

$$\int_Q e^{2s(\delta-1)\alpha}(u^2 + u_t^2) \leq C_3 \|y_0\|_{L^2}^2.$$

We are now ready to apply Kakutani's theorem. Consider $\Phi : K \rightarrow 2^K$,

$$(4.8) \quad \begin{aligned} \Phi(\tilde{y}) = \{y \mid & y \in K, y(T) \equiv 0, \text{ and } \exists u \in H^1([0, T]; L^2(I)), \\ & \text{with } \int_Q e^{2s(\delta-1)\alpha}(u^2 + u_t^2) \leq C_3 \|y_0\|_{L^2}^2, \\ & \text{such that } (u, y) \text{ satisfy (1.2)} \} \end{aligned}$$

Clearly Φ is well-defined, takes non-empty values for every $\tilde{y} \in K$, and has convex values.

In order to prove that Φ has closed values, first let us fix a $\tilde{y}$, and consider a convergent sequence

$$y_n \rightarrow y \text{ in the } L^2(Q) \text{ norm,}$$

with $y_n \in \Phi(\tilde{y})$, and choose the corresponding u_n as in the definition of Φ above (4.8). Since $y_n \in K$, $\forall n$, and K is compact, it follows that their limit y is also in K . Furthermore,

$$\begin{aligned} \|y_n(T) - y(T)\|_{L^2(I)} &\leq \|y_n - y\|_{C([0, T]; L^2(I))} \\ &\leq \|y_n - y\|_{H^1([0, T]; L^2(I))}^{1/2} \|y_n - y\|_{L^2(Q)}^{1/2} \\ &\leq C \rho^{1/2} \cdot \|y_n - y\|_{L^2(Q)}^{1/2}, \end{aligned}$$

so $y(T) \equiv 0$.

Since $\int_Q e^{2s(\delta-1)\alpha}(u^2 + u_t^2) \leq C_3 \|y_0\|_{L^2(I)}^2$, we may select a weakly convergent subsequence which tends to a limit u with $\int_Q e^{2s(\delta-1)\alpha}(u^2 + u_t^2) \leq C_3 \|y_0\|_{L^2(I)}^2$ (the proof is as above). It can be seen that $(y_n)_t \rightarrow y_t$ strongly in the $H^{-1}(Q)$ norm and $(b \cdot (y_n)_x)_x \rightarrow (by_x)_x$ strongly in the $L^2((0, T); H^{-2}(I))$ norm. By passing to the weak limit in equation (1.2), satisfied by (u_n, y_n) , we obtain that (u, y) also satisfy the equation. Therefore $y \in \Phi(\tilde{y})$, and thus $\Phi(\tilde{y})$ is closed; in fact, since $\Phi(\tilde{y}) \subset K$, it follows that $\Phi(\tilde{y})$ is compact for every $\tilde{y} \in K$.

In this case, the lower semicontinuity of Φ can be obtained from the fact that it has a closed graph. Indeed, if $\tilde{y}_n \in K$, $\tilde{y}_n \rightarrow \tilde{y}$, and $y_n \in \Phi(\tilde{y}_n) \rightarrow y$ in the $L^2(Q)$ norm, consider the corresponding u_n , as above, and we obtain (by taking a subsequence) that

$$\begin{aligned} u_n &\rightarrow u && \text{weakly in } L^2(Q), \\ (u_n)_t &\rightarrow u_t && \text{weakly in } L^2(Q), \text{ with } \int_Q e^{2s(\delta-1)\alpha}(u^2 + u_t^2) \leq C_3 \|y_0\|_{L^2}^2; \\ (y_n)_t &\rightarrow y_t && \text{strongly in } H^{-1}(Q); \\ (y_n)_x &\rightarrow y_x && \text{strongly in } L^2((0, T); H^{-1}(I)); \\ \tilde{y}_n &\rightarrow \tilde{y} && \text{strongly in } L^2(Q). \end{aligned}$$

Since $|a'(\tilde{y}_n) - a'(\tilde{y})| \leq C|y_n - y|$, it also follows that $a'(\tilde{y}_n) \rightarrow a'(\tilde{y})$ strongly in $L^2(Q)$ (and weakly-star in $L^\infty(Q)$). Then $a'(\tilde{y}_n)(y_n)_x \rightarrow a'(\tilde{y})y_x$ weakly in $L^2((0, T); H^{-1}(I))$, or, equivalently, $(a'(\tilde{y}_n)(y_n)_x)_x \rightarrow (a'(\tilde{y})y_x)_x$ weakly in $L^2((0, T); H^{-2}(I))$. By going to the weak limit in equation (1.2) we obtain that the pair (u, y) also satisfies the linearized system, with $b = a'(\tilde{y})$. The other conditions ($\tilde{y}, y \in K$, $y(T) \equiv 0$) are obviously satisfied (see above for proof details), so $(\tilde{y}, y)$ belongs to the graph of Φ .

Thus we can apply Kakutani's theorem and obtain that there is $y \in K$ such that $y = \Phi(y)$. Such a y is a solution of the diffusion equation (1.1) with $y(0) = y_0$ and $y(T) \equiv 0$. In addition, its controller u satisfies the required estimate

$$\int_Q e^{2s(\delta-1)\alpha}(u^2 + u_t^2) \leq C_3 \|y_0\|_{L^2}^2.$$

Now let us come back to the more general case, $y_0 \in H^1(I)$.

Theorem 4.2 *For any $\delta > 0$ there exists $\eta > 0$ such that, for every $y_0 \in H^1(I)$ with $\|y_0\|_{H^1(I)} \leq \eta$, the equation (1.1) is exactly null controllable, with a controller u satisfying*

$$(4.9) \quad \int_Q e^{2s(\delta-1)\alpha}(u^2 + u_t^2) \leq C_3 \|y_0\|_{L^2}^2.$$

Proof of Theorem 4.2 We are going to divide the interval $[0, T]$ into two parts, by choosing $0 < T_0 < T$. On the first part of the interval we shall make $u \equiv 0$; the equation becomes

$$(4.10) \quad \begin{aligned} y_t - (by_x)_x &= 0 & \forall (x, t) \in Q \\ y(x, t) &= 0 & \forall (x, t) \in \Sigma \\ y(x, 0) &= y_0 & x \in I, \end{aligned}$$

where we have renamed $I \times [0, T_0] \equiv Q_{T_0} \equiv Q$ and $\partial I \times [0, T_0] \equiv \Sigma$ for greater convenience. We shall use the regularizing properties of equation (4.10), eventually obtaining that

$$(4.11) \quad \int_I (a(y))_{xx}^2(t) + y_x^2(t) + y^2(t) \, dx \leq C \int_I y_0^2 + y_x^2(0) \, dx$$

Then, applying the previous Theorem 4.1, we shall be able to establish the null controllability of y on the interval $[T_0, T]$. Finally, let us note that the function u defined by

$$u(x, t) = \begin{cases} 0, & \text{for } t \in [0, T_0] \\ u^*(x, t), & \text{for } t \in [T_0, T] \end{cases},$$

where u^* is a controller for y on $[T_0, T]$, still has all the desired properties (it belongs to $H^1([0, T]; L^2(I))$ and it satisfies (4.9)). Then the proof will be complete.

Now let us establish estimate (4.11). For the beginning, we shall only assume in (4.10) that $b_t \in L^2(Q)$ and $b_x \in L^\infty([0, T_0]; L^2(I))$.

Multiplying equation (4.10) by y and integrating on Q_t we get

$$(4.12) \quad \int_I y^2(t) \, dx + 2 \int_{Q_t} by_x^2 = \int_I y^2(0) \, dx.$$

Multiplying the same equation by y_t and integrating by parts for t we get

$$\int_{Q_t} y_t^2 + \frac{1}{2} \int_I by_x^2(t) = \frac{1}{2} \int_I by_x^2(0) + \frac{1}{2} \int_{Q_t} b_t y_x^2.$$

Then, it is true that

$$\int_{Q_t} y_t^2 + \int_I y_x^2(t) \, dx \leq C \left(\int_I y_x^2(0) \, dx + \left(\int_{Q_t} b_t^2 \right)^{\frac{1}{2}} \cdot \left(\int_{Q_t} y_x^4 \right)^{\frac{1}{2}} \right),$$

and, consequently, (since $0 \leq t \leq T_0$)

$$\begin{aligned} \int_Q y_t^2 + \sup_{t \in [0, T_0]} \int_I y_x^2(t) \, dx &\leq C \left(\int_I y_x^2(0) \, dx + \right. \\ &\quad \left. + \|b_t\|_{L^2(Q)} \cdot \|y_x\|_{L^2((0, T_0); L^\infty(I))} \cdot \sup_{t \in [0, T_0]} \|y_x(t)\|_{L^2(I)} \right). \end{aligned}$$

Let us note that $\|y_x\|_{L^2((0,T_0);L^\infty(I))} \leq C\|(by_x)_x\|_{L^2(Q)} = C\|y_t\|_{L^2(Q)}$, so

$$\begin{aligned} \int_Q y_t^2 + \sup_{t \in [0, T_0]} \int_I y_x^2(t) \, dx &\leq C_5 \left(\int_I y_x^2(0) \, dx + \right. \\ &\quad \left. + \|b_t\|_{L^2(Q)} \left(\int_Q y_t^2 + \sup_{t \in [0, T_0]} \int_I y_x^2(t) \, dx \right) \right) \end{aligned}$$

Making $\|b_t\|_{L^2(Q)}$ sufficiently small ($\|b_t\|_{L^2(Q)} \leq \frac{1}{2C_5}$, for example), we obtain

$$\int_Q y_t^2 + \sup_{t \in [0, T_0]} \int_I y_x^2(t) \, dx \leq C \int_I y_x^2(0) \, dx.$$

As an immediate consequence, we also get $\int_Q (by_x)_x^2 \leq C \int_I y_x^2(0)$ (this will be useful later on).

Rewriting the last inequality (and remembering that, by definition, $b = a'(\tilde{y})$), we have obtained

$$\|y_t\|_{L^2(Q)} + \sup_{t \in [0, T_0]} \|y_x(t)\|_{L^2(I)} \leq C_6 \|y_x(0)\|_{L^2(I)},$$

if $\|\tilde{y}_t\|_{L^2(Q)} \leq \frac{1}{2MC_5}$.

Consider the set

$$(4.13) \quad K = \{y \mid \|y_t\|_{L^2(Q)} + \sup_{t \in [0, T_0]} \|y_x(t)\|_{L^2(I)} \leq \frac{1}{2MC_5}, y(0) = y_0\},$$

which is clearly compact in $L^2(Q)$ (being closed in $L^2(Q)$ and bounded in $H^1(Q)$). Then, define $\Phi : K \rightarrow 2^K$ by

$$(4.14) \quad \Phi(\tilde{y}) = \{y \mid y \in K, y \text{ is a solution of equation (4.10)}\}.$$

For $\|y_x(0)\|_{L^2(I)} \leq \frac{1}{C_6 \cdot 2MC_5}$, the multifunction Φ has non-empty values everywhere. It can be easily checked that it also has convex values.

Consider a sequence of solutions $y_n \in \Phi(\tilde{y})$ which converges in $L^2(Q)$ to a limit y . Then $(y_n)_t \rightarrow y_t$ in $H^{-1}(Q)$ and $(b \cdot (y_n)_x)_x \rightarrow (by_x)_x$ in $L^2((0, T_0); H^{-2}(I))$. By going to the limit in equation (4.10) for y_n we obtain that $y_t - (by_x)_x = 0$ in the weak sense (in $H^{-2}(Q)$), and therefore $y \in \Phi(\tilde{y})$. Thus we have proved that Φ has closed values (compact, as a matter of fact).

All that is left is to show that Φ has a closed graph (and thus is upper semicontinuous). Indeed, if we take two sequences y_n and $\tilde{y}_n$, $y_n \in \Phi(\tilde{y}_n)$,

converging to y and to $\tilde{y}$ respectively, we successively obtain that

$$\begin{aligned}
(y_n)_t &\rightarrow y_t && \text{in } H^{-1}(Q); \\
(y_n)_x &\rightarrow y_x && \text{in } L^2((0, T_0); H^{-1}(I)); \\
\tilde{y}_n &\rightarrow \tilde{y} && \text{in } L^2(Q); \\
a'(\tilde{y}_n) &\rightarrow a'(\tilde{y}) && \text{weakly-star in } L^\infty(Q); \\
a'(\tilde{y}_n) \cdot (y_n)_x &\rightarrow a'(\tilde{y})y_x && \text{in } L^2((0, T_0); H^{-1}(I)); \\
(a'(\tilde{y}_n) \cdot (y_n)_x)_x &\rightarrow (a'(\tilde{y})y_x)_x && \text{in } L^2((0, T_0); H^{-2}(I)).
\end{aligned}$$

By going to the limit in equation (4.10), satisfied by all the functions y_n with $b = a'(\tilde{y}_n)$, we have obtained that $y \in \Phi(\tilde{y})$, so the graph is closed.

Now all the conditions needed to apply Kakutani's theorem are fulfilled and we obtain that Φ has a fixed point $y \in \Phi(y)$, and therefore the diffusion equation

$$\begin{aligned}
(4.15) \quad & y_t - (a(y))_{xx} = 0 \quad \forall (x, t) \in Q = I \times (0, T_0) \\
& y(x, t) = 0 \quad \forall (x, t) \in \Sigma = \partial I \times (0, T_0) \\
& y(x, 0) = y_0 \quad x \in I,
\end{aligned}$$

has a solution y with

$$(4.16) \quad \|y_t\|_{L^2(Q)} + \sup_{t \in [0, T_0]} \|y_x(t)\|_{L^2(I)} \leq \underbrace{\frac{1}{2MC_5}}_C \|y_x(0)\|_{L^2(I)},$$

for $\|y_x(0)\|_{L^2(I)} \leq \frac{1}{C_6 \cdot 2MC_5}$.

However, the solution of equation (4.15) has further regularity properties, of which we are going to need only one. For convenience, we shall keep the notation $b = a'(\tilde{y}) \equiv a'(y)$. By multiplying equation (3.1) with $(by_x)_{xt}$, we get

$$\int_{Q_t} (by_x)_x (by_x)_{xt} + \int_{Q_t} by_{xt}^2 + \int_{Q_t} y_{xt} b_t y_x = 0,$$

hence

$$\frac{1}{2} \int_I t (by_x)_x^2(t) dx + \int_{Q_t} t by_{xt}^2 + \int_{Q_t} t y_{xt} b_t y_x = \frac{1}{2} \int_{Q_t} (by_x)_x^2.$$

In this equation we have

$$\left| \int_{Q_t} t y_{xt} b_t y_x \right| \leq \frac{\mu}{2} \int_{Q_t} t y_{xt}^2 + \frac{1}{2\mu} \int_{Q_t} t b_t^2 y_x^2,$$

where we may take μ to be the same constant as in the beginning of the paper ($\mu \leq a'(x) \forall x \in \mathbb{R}$, so that $\mu \leq a'(y) = b$). Then,

$$\begin{aligned} \int_{Q_t} t b_t^2 y_x^2 &\leq \int_Q b_t^2 \cdot \|t y_x^2\|_{L^\infty(Q)} \\ &\leq C \int_Q y_t^2 \cdot \sup_{t \in [0, T_0]} \int_I t (b y_x)_x^2(t) \, dx. \end{aligned}$$

Consequently, we obtain that

$$\sup_{t \in [0, T_0]} \int_I t (b y_x)_x^2(t) \, dx + \int_Q t y_{xt}^2 \leq C_7 \left(1 + \sup_{t \in [0, T_0]} \int_I t (b y_x)_x^2(t) \, dx \right) \int_I y_x^2(0) \, dx,$$

$\forall t \in [0, T_0]$. Again, for sufficiently small $\|y_x^2(0)\|_{L^2(I)}$ (such as $\|y_x^2(0)\|_{L^2(I)}^2 \leq \frac{1}{2C_7}$), we have proved that

$$(4.17) \quad \sup_{t \in [0, T_0]} \int_I t (b y_x)_x^2(t) \, dx + \int_Q t y_{xt}^2 \leq C \int_I y_x^2(0) \, dx.$$

By choosing $t = T_0$ in (4.12), (4.16), and (4.17), we obtain

$$\int_I (a(y))_{xx}^2(t) + y_x^2(t) \, dx \leq C \left(\int_I y_0^2 \, dx + \int_I y_x^2(0) \, dx \right)$$

for any y_0 with $\|y_x(0)\|_{L^2(I)}$ sufficiently small. This concludes the proof of Theorem 4.2.

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